

```

import torch
import pandas as pd
import re
from datasets import load_dataset
from transformers import AutoTokenizer, AutoModelForCausalLM

print("Downloading GSM8K...")
ds = load_dataset("gsm8k", "main")
data = ds["train"].to_pandas()
print("Dataset size:", len(data))

device = "cuda" if torch.cuda.is_available() else "cpu"
print("Using:", device)

model_name = "TinyLlama/TinyLlama-1.1B-Chat-v1.0"

tokenizer = AutoTokenizer.from_pretrained(model_name)

model = AutoModelForCausalLM.from_pretrained(
    model_name,
    torch_dtype=torch.float16 if device=="cuda" else torch.float32
).to(device)

model.eval()

def build_prompt(question, mode="zero_shot", examples=None):

    if mode == "zero_shot":
        return f"Solve the math problem. Give only number.\n\nQuestion: {question}\nAnswer:"

    elif mode == "few_shot":
        p = "Examples:\n\n"
        for q,a in examples:
            p += f"Q: {q}\nA: {a}\n\n"
        p += f"Q: {question}\nA:"
        return p

    elif mode == "cot":
        return f"Solve step by step.\nQuestion: {question}\nLet's think step by step."

def generate_answer(prompt):

    inputs = tokenizer(prompt, return_tensors="pt").to(device)

    with torch.no_grad():
        out = model.generate(
            **inputs,
            max_new_tokens=64,
            do_sample=True,
            temperature=0.7
        )

    return tokenizer.decode(out[0], skip_special_tokens=True)

def extract_last_number(text):
    nums = re.findall(r"[-+]?\d*\.\d+|\d+", text)
    return nums[-1] if nums else None

def get_examples(k=3):
    ex = []
    for i in range(k):
        q = data.iloc[i]["question"]
        a = data.iloc[i]["answer"].split("####")[-1].strip()
        ex.append((q,a))
    return ex

few_examples = get_examples(3)

def evaluate(mode, n=10):

    correct = 0
    hall = 0

    for i in range(n):

```

```
q = data.iloc[i]["question"]
gt = data.iloc[i]["answer"].split("####")[-1].strip()

if mode == "few_shot":
    prompt = build_prompt(q, mode, few_examples)
else:
    prompt = build_prompt(q, mode)

out = generate_answer(prompt)

pred = extract_last_number(out)
gt = extract_last_number(gt)

if pred is None:
    hall += 1
elif pred == gt:
    correct += 1

print(mode, "processed", i+1, "/", n)

acc = correct / n
hall_rate = hall / n

print("\nMode:", mode)
print("Accuracy:", acc)
print("Hallucination:", hall_rate)

return acc, hall_rate

acc_zero, h_zero = evaluate("zero_shot", 10)
acc_few, h_few = evaluate("few_shot", 10)
acc_cot, h_cot = evaluate("cot", 10)

!pip install -q transformers torch accelerate pandas pyarrow datasets
```

```

Downloading GSM8K...
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
README.md: 7.93k/? [00:00<00:00, 505kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
main/train-00000-of-00001.parquet: 100% 2.31M/2.31M [00:02<00:00, 2.65MB/s]
main/test-00000-of-00001.parquet: 100% 419k/419k [00:00<00:00, 973kB/s]
Generating train split: 100% 7473/7473 [00:00<00:00, 81365.52 examples/s]
Generating test split: 100% 1319/1319 [00:00<00:00, 41226.35 examples/s]
Dataset size: 7473
Using: cpu
config.json: 100% 608/608 [00:00<00:00, 57.9kB/s]
tokenizer_config.json: 1.29k/? [00:00<00:00, 110kB/s]
tokenizer.json: 1.84M/? [00:00<00:00, 54.1MB/s]
tokenizer.model: 100% 500k/500k [00:00<00:00, 680kB/s]
special_tokens_map.json: 100% 551/551 [00:00<00:00, 43.1kB/s]
`torch_dtype` is deprecated! Use `dtype` instead!
model.safetensors: 100% 2.20G/2.20G [00:19<00:00, 118MB/s]

```

```

Loading weights: 100% 201/201 [00:03<00:00, 64.35it/s, Materializing param=model.norm.weight]
https://colab.research.google.com/drive/1U8rgWLba4IO-HQd-1uOfWygS WjdROEiV?usp=sharing
generation_config.json: 100% 124/124 [00:00<00:00, 12.7kB/s]

```

```

EXP 5
zero_shot processed 1 / 10
zero_shot processed 2 / 10
zero_shot processed 3 / 10
zero_shot processed 4 / 10
zero_shot processed 5 / 10
zero_shot processed 6 / 10
zero_shot processed 7 / 10
zero_shot processed 8 / 10
zero_shot processed 9 / 10
zero_shot processed 10 / 10

```

```

Mode: zero_shot
Accuracy: 0.0
Hallucination: 0.0
few_shot processed 1 / 10
few_shot processed 2 / 10
few_shot processed 3 / 10
few_shot processed 4 / 10
few_shot processed 5 / 10
few_shot processed 6 / 10
few_shot processed 7 / 10
few_shot processed 8 / 10
few_shot processed 9 / 10
few_shot processed 10 / 10

```

```

Mode: few_shot

```